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Sourceless Logging While Drilling Platform Enables Reservoir Evaluation in Complex and High-Risk Drilling Environment: A Case Study from Abdali and Ahmadi Field, Kuwait

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Abstract

This study focuses on the deployment of a sourceless formation evaluation Logging-While-Drilling (LWD) platform in the Ahmadi and Abdali fields, operated by Kuwait Oil Company (KOC). These fields primarily produce light to medium crude oil from clastic and carbonate reservoirs. Due to geological complexity and field location, drilling operations commonly involve high-angle and horizontal wells, with associated challenges such as differential sticking and the risk of losing in hole the bottomhole assembly (BHA). The implementation of the sourceless LWD platform mitigates the risk of radioactive source loss in the wellbore and eliminates the complex abandonment procedure, enhancing operational safety and efficiency.

The sourceless LWD platform was deployed for the complete formation evaluation in two fields in Kuwait. The LWD tool utilizes a pulsed neutron generator (PNG) and delivers a suite of advanced formation evaluation measurements, including bulk density, neutron porosity, elemental capture spectroscopy and sigma. In addition, it provides gamma ray and electromagnetic propagation resistivities for comprehensive reservoir characterization. The platform also includes ultrasonic caliper and annular pressure while drilling measurements for drilling safely.

In the Abdali field, the logging was conducted in washdown mode to minimize operational risk in unstable formations, reduce the likelihood of BHA sticking, and enable a faster penetration rate through low-pore pressure zones while still acquiring essential data for formation evaluation. In the Ahmadi field, the sourceless LWD platform was part of the drilling BHA for the long ERD well with a tight pore pressure gradient.

Data acquisition was completed in both wells, delivering the full suite of measurements required to characterize the target reservoirs and support potential economic evaluation. The deployment of the sourceless LWD platform demonstrated significant HSE benefits across all stages of operation - including tool handling, transportation, wellsite operations, and potential abandonment - by eliminating the risk associated with radioactive sources.

The formation evaluation results in the two wells showed good comparison with the offset wells logged using chemical sources. This enabled the successful evaluation of the Ratawi limestone in the first ERD well in Abdali field and the Wara formation along the ERD well in Ahmadi field. Moreover, the ability to put technology close to the bit, together with high-speed telemetry, enabled faster logging speed in washdown and minimized rathole for drilling operations.

A novel sourceless neutron-gamma density has been introduced to derive bulk density measurements without the use of chemical radioactive sources. This new sourceless density expands the applicability across a broader range of reservoirs and drilling conditions, while significantly reducing Health, Safety and Environmental (HSE) risk.

Introduction

The Lower Cretaceous reservoirs are among the most productive in Kuwait, contributing significantly to oil output across various structural traps. These reservoirs, particularly prominent in Kuwait's onshore fields, are well known for their complex geological characteristics and heterogeneity. This study focuses on Well 1 from the Abdali field in North Kuwait and Well 2 from the Ahmadi field in the Southeast (Figure 1).

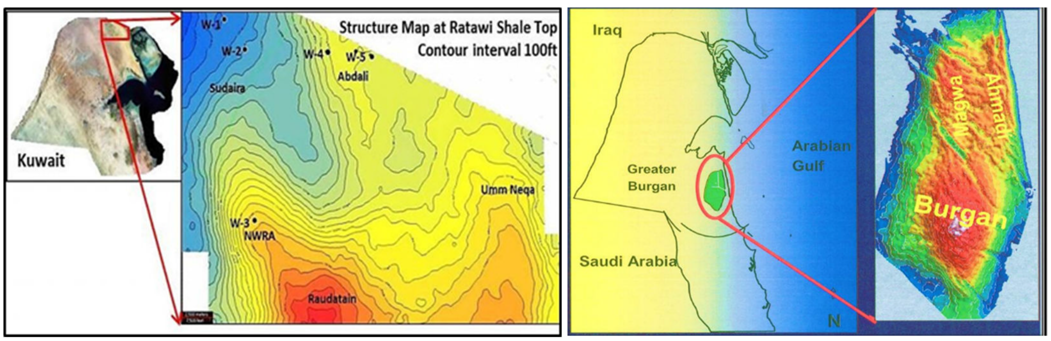

Figure 1—(left) Location of Abdali field; (right) Location of Ahmadi field.

Since 2005, horizontal and extended-reach drilling (ERD) has been increasingly adopted in these fields to improve reservoir contact, enhance drainage, and mitigate challenges such as water coning and gas breakthrough, especially in layered carbonate and depleted clastic formations. While early production was driven by thick, massive sand bodies, current development efforts target more heterogeneous and thinner deltaic sands, requiring advanced drilling and evaluation strategies.

The Abdali and Ahmadi fields each present unique reservoir-specific drilling risks. Abdali's stacked reservoirs, such as the Zubair and Ratawi formations, pose challenges related to stuck pipe incidents, wellbore instability, fractured intervals, and low reservoir pressure. In contrast, the Ahmadi field, which encompasses the Wara, Ahmadi, and Burgan formations, is characterized by tight pore pressure gradients and unstable shale zones.

To address these risks, this study demonstrates the successful application of a sourceless Logging-While-Drilling (LWD) platform. Deployed in washdown mode in Abdali field and in drilling mode in Ahmadi field, the platform reduced operational hazards while maintaining data quality. Real-time formation evaluation, enabled by high-speed telemetry and strategic tool placement near the bit, allowed efficient navigation through geologically challenging intervals, and ensured acquisition of essential reservoir data.

Risks and HSE Challenges in High-Angle Drilling

Over the past two decades, the oil industry has shifted most of the development drilling activities to high-angle and/or horizontal wells to maximize reservoir contact and increase hydrocarbon production. However, this transition introduces additional risks due to well geometry, increased torque and drag, and the likelihood of getting stuck (Al-Rubaiyea et. al, 2015). Drilling risks associated with unexpected pore pressure ramps, depleted reservoirs, and other geohazards can increase the probability of a BHA becoming stuck and, in more severe cases, lost in the hole.

Key challenges in high-angle drilling include:

- Hole Cleaning Issues: Cuttings tend to settle on the lower side of the hole, making it harder to transport out. Poor hole cleaning can lead to stuck pipe (mechanical stuck), pack-offs, or increased torque and drag.
- Stuck Pipe Risks: Differential sticking risk is higher due to the continuous and increased contact of the BHA with the borehole wall.

In summary, drilling high-angle and horizontal wells involves increased operational complexity, greater mechanical risk, and higher costs. It requires careful planning, advanced technology, and constant monitoring to mitigate these risks.

Two of the most critical measurements used for formation evaluation are bulk density and neutron porosity. Historically, radioactive chemical sources have been used in the logging tools to derive these measurements. These sources are classified as hazardous materials. Direct radiation exposure can cause serious health issues like radiation burns, cancer, etc. Any worker handling these sources must follow strict safety protocols. When these sources are being transported to and from drilling rigs, they are stored in shields that protect personnel from exposure. The chemical radioactive sources used in logging tools can pose health, environmental, and security risks if not responsibly managed.

Most of the nuclear logging tools in the industry use two distinct types of natural radioactive sources:

a. Cesium gamma ray source used to measure the formation density with a half-life period of approximately 30 years.
b. Americium-Beryllium neutron source used to measure the formation Hydrogen Index with a half-life period of more than 400 years.

When a BHA having radioactive sources gets stuck in the hole, multiple attempts must be made to retrieve the tools. If the attempts fail, then the chemical sources are tried to be fished from the tool (this capability only exists in certain tools for certain vendors). If the fishing fails, then specific procedures are followed to safely abandon the chemical source in the ground. The country's nuclear authority regulates the abandonment procedures. A general summary to minimize damage to the local environment would be:

- The tool having the radioactive sources must be cemented downhole using special cement. (Al-Rubaiyea et. al, 2015)
- Mud return should be watched to ensure there is no mud contamination with radiation.
- The loss of the sources should be properly documented with the country authorities.
- A certain area around the well trajectory must be avoided in future drilling, leading to loss of recoverable oil.
- Future oil production at the gathering station should be monitored to detect any abnormal oil radioactivity.

Sourceless Logging platform

The new sourceless formation evaluation LWD tool is a multifunction tool that combines multiple formation evaluation measurements, annular pressure while drilling (APWD) and ultrasonic caliper measurements in a single collar (Figure 2). At the heart of the LWD tool is a pulsed neutron generator (PNG), which eliminates the need for an americium beryllium (AmBe) chemical neutron source. The high-energy neutrons interact with the formation and fluids in a few ways to either slow down, get absorbed, or generate gamma rays. All these mechanisms are measured by several detectors located in the tool to determine key formation properties. These include elemental capture spectroscopy for detailed lithology, formation capture cross section (sigma) for water saturation, neutron porosity and neutron gamma density (NGD*) essential for formation evaluation. The use of PNG in the LWD eliminates the need for a chemical source to measure the nuclear properties of the formation.

Figure 2—Tool layout and measurement configuration.

Sourceless Density Measurement

In the sourceless density measurement, a pulsed neutron generator or PNG replaces the traditional cesium-based gamma-gamma density source. PNG has been in use by industry for many decades, first in wireline tools, and almost two decades ago, in logging-while-drilling tools (Weller et al., 2005). PNG generates 14-MeV neutrons, many of which interact with the nuclei in the formation through inelastic collisions. These interactions produce inelastic gamma rays, which can be distinguished from other gamma signals based on their arrival time, thanks to the pulsed nature of the neutron source. These inelastic gamma rays are shown as the green curve in Figure 3 (Reichel et al, 2012).

Unlike traditional source-based density tools, the inelastic gamma count rate in sourceless density measurements is not directly or monotonically related to formation density. This is because the gamma rays originate from a secondary source—the neutron-induced interactions within the formation—rather than from a fixed-point source. The spatial distribution of this secondary source depends on neutron transport, illustrated by the blue curve in Figure 3 and it influences the number of gamma rays that return to the detector.

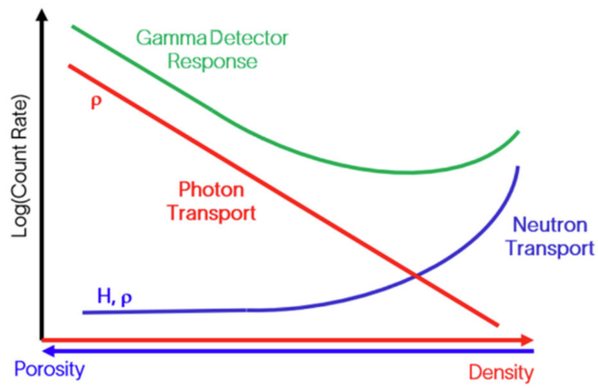

Figure 3—Response of the gamma ray detector is a function of the neutron and gamma transport and attenuation.

To extract a meaningful density measurement, the inelastic gamma count rate must be corrected for neutron transport effects. After this correction, the remaining signal reflects gamma ray attenuation due to Compton scattering, which is related to the formation's electron density, similar to the principle behind traditional source-based density tools. This corrected response is represented by the red curve in Figure 3.

The nuclear measurements are susceptible to the drilling environment. Neutron and photon transport is affected by mud properties such as density and salinity, and hole size (Mauborgne et al., 2024). Moreover, neutron gamma density is not borehole compensated. Thus, precise environmental corrections are required to obtain accurate data. The hole size is determined using an ultrasonic caliper, which requires calibration of the mud slowness inside the casing. This calibration becomes even more important for larger boreholes of 9-inch and greater.

The NGD measurement is also affected by the presence of heavy elements, such as iron, in the formation. Heavy elements have a higher probability of capturing a neutron and emitting a larger number of gamma rays compared to lighter elements present in the reservoir rock. To mitigate this, corrections are made using information from the spectroscopy and sigma measurements.

Sourceless Density Considerations

It is important to consider the depth of investigation (DOI) and vertical resolution (VR) of the sourceless neutron-gamma density when comparing or using it in place of the gamma-gamma density. Due to neutron transport, the volume of investigation of the sourceless density is much larger. This results in larger vertical resolution and deeper depth of investigation, making the measurement less sensitive to invasion and thin layers.

It is challenging to successfully focus high-energy neutrons generated by PNG in a narrow beam to enable azimuthal measurements, such as a borehole density image. So, when running the sourceless platform, the operators will not have access to a density image, but natural gamma ray and ultrasonic caliper images will still be available for geosteering and wellbore stability applications, respectively.

Case Studies

Abdali field

The Abdali field in North Kuwait has multiple stacked reservoir zones in the Zubair and Ratawi formations. Drilling challenges for developing these highly depleted reservoirs include stuck pipe incidents that often lead to side-tracking. The problems are observed in the Shuaiba and Ahmadi overburden formations, as well as in the reservoir target zones. Specifically, the intersection of shale layers during drilling triggers mechanical and time-dependent wellbore stability issues. Additionally, intersecting fractured zones presents issues such as mud losses (Al-Hamad, et. al, 2023).

In the Abdali field, LWD logging for well 1 was conducted in washdown mode post-drilling to minimize operational risk in unstable formations and the likelihood of BHA sticking. Washdown logging is done at a much faster rate than drilling.

Well 1

Well 1 was the first extended-reach drilling (ERD) well in Abdali field. The LWD sourceless platform was deployed in a 9.25-inch section drilled with oil-based mud with a density of 11.6 ppg. This section covered an interval of 2600 ft with a maximum deviation of 55 degrees, crossing a sequence of shale and sandstone before reaching the shale and limestone formations (Figure 4).

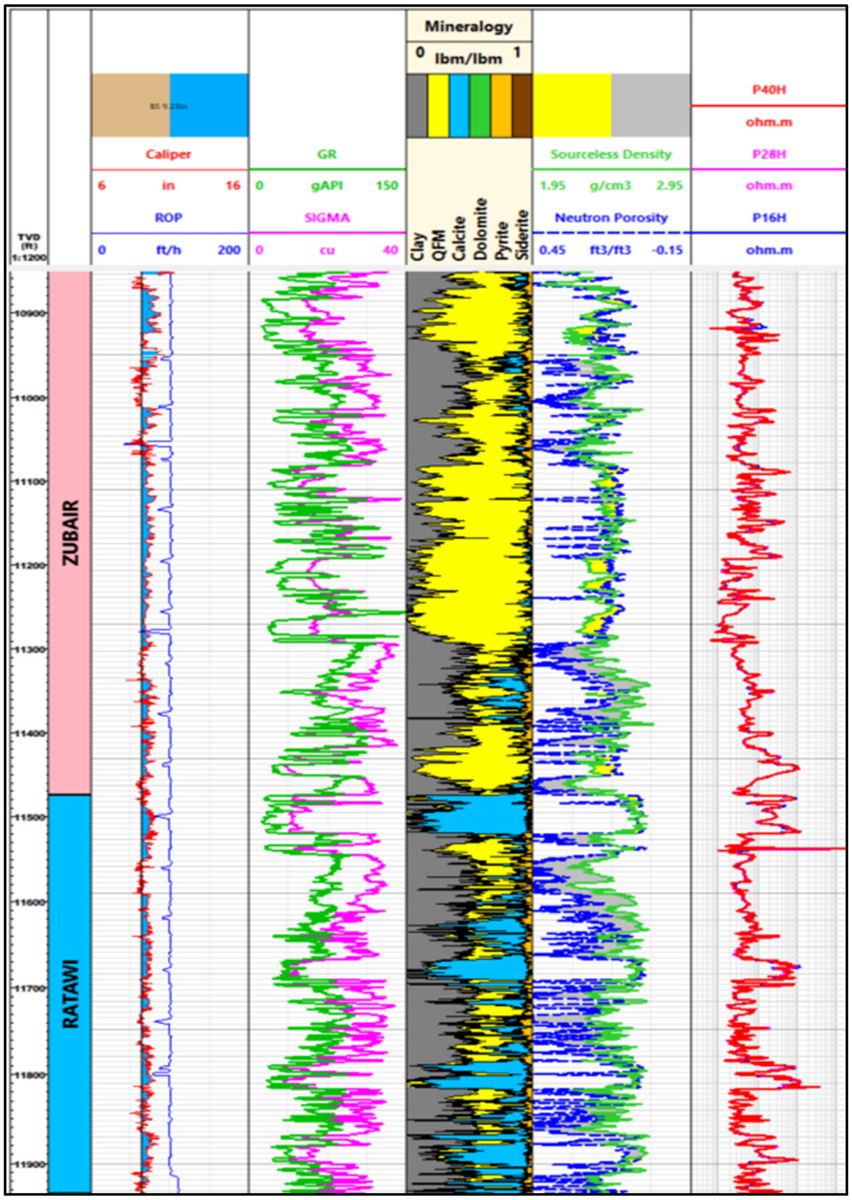

Figure 4—Overview of well 1 illustrating the Zubair and Ratawi Shale formations. Looking at the log data tracks starting from the left track to the right, track 1 provides hole shape indication with caliper and logging speed. In track 2, gamma ray and sigma, while in track 3, the mineralogy derived from elemental fractions from spectroscopy is shown. Track 4 has the sourceless density (SLRO) together with neutron porosity (TNPH). Track 5 shows the propagation resistivity.

The BHA passed through Zubair Formation and Ratawi shale, enabling access to the well's main target, the Ratawi limestone, which was drilled in a 6-inch borehole after this section.

The Zubair Formation consists of clastic sandstone interbedded with gray to black, thinly laminated siltstone and shale. Sigma and GR measurements show the changes in clay content within the logged interval, as in Figure 4. The lithology of Zubair formation is further confirmed by spectroscopy measurements obtained from the sourceless logging platform, which show a strong correlation with the mud log. The shale layer within the Zubair formation serves as an effective seal for the oil and gas reservoirs in the interbedded Zubair sand units.

The LWD mineralogy from spectroscopy measurements accurately identified the top of Ratawi formation. The Ratawi formation is divided into an upper Ratawi Shale member and a lower Ratawi Limestone member (Ratawi Oolite), reflecting the proportions of lime mud-wackestone, calcareous shales, and marls in each unit. This highlights the complexity of mineralogy, which could pose a challenge for petrophysical interpretation if spectroscopy is not available.

A correlation with an offset well was performed to confirm the consistency of reservoir properties. Figure 5 shows the correlation between Well 1, logged with the LWD sourceless logging platform, and a vertical offset well logged using a wireline triple combo (GR, caliper, resistivity, and density-neutron). There is clear similarity and consistency across the various logs in the two wells. Please note the effect of shale breakouts in zone B in an already large hole on the density and neutron measurement in well 1. These breakouts are affecting neutron porosity and density because the tool is measuring the total response of the formation and fluid filled in deep directional breakouts in its volume of investigation. The ultrasonic signal gets severely attenuated when traveling in azimuthal sectors with the larger standoff. The caliper value averages only over the sectors with a valid response, underestimating the hole size, leading to under correction of density and neutron porosity for the hole size. This is causing neutron porosity to read high and density low in these shale intervals.

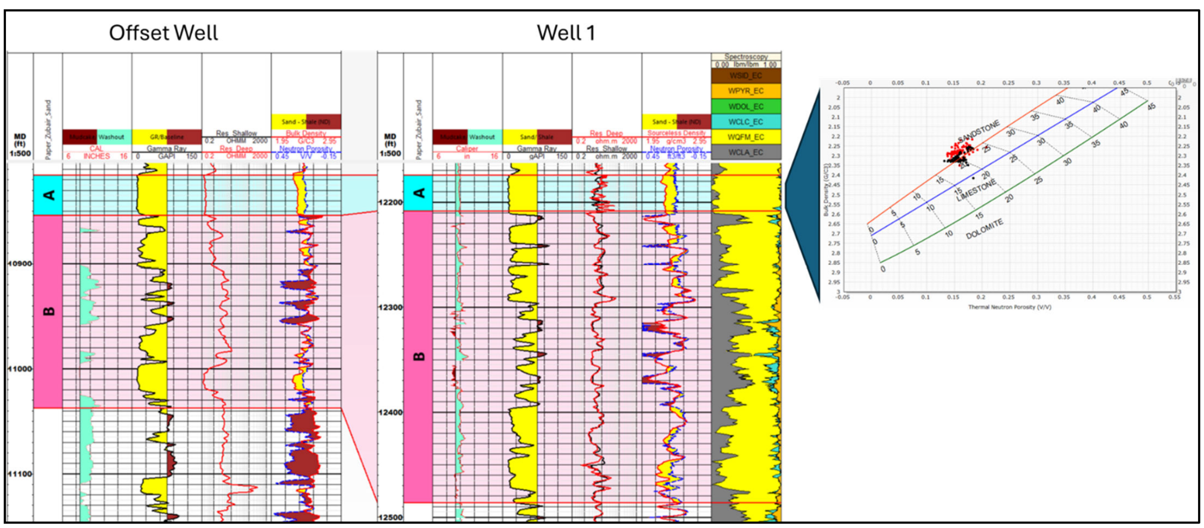

Figure 5—Left -Well correlation between the offset well and Well 1 through Sand A and Sand B in the Zubair formation. Right - The density-neutron crossplot displays Sand A from both wells and confirms the presence of light oil in Well 1.

The density-neutron crossplot from the two wells, well 1 (red) and the offset well (black), in Sand A shows that the porosity range is similar in this clean sand. However, the difference in density is consistent with the higher resistivity in Well 1, indicating higher oil saturation in this layer compared to the offset well.

Ahmadi field

The Ahmadi field, where the Ahmadi, Wara, Mauddud and Burgan formations are the reservoirs of interest, is characterized by tight pore pressure gradients (differential pressure in the reactive/ stressed shales vs the depletion of the producing sandstone and carbonate reservoirs) (Hussein et. al., 2013). The transgressive

Ahmadi Formation was deposited during the Early Cretaceous. The Lower Ahmadi carbonate is interbedded between Ahmadi shale at the top and Wara shale below. In the Ahmadi field, the sourceless LWD platform was deployed as part of the drilling BHA for the long ERD Well 2.

Well 2

Well 2 was a long extended-reach drilling (ERD) well. The LWD sourceless platform was deployed in the 8.5-inch borehole section, drilled with oil-based mud with a density of 13.2 ppg. This section covered an interval of 2480 ft with a maximum deviation of 75 degrees, and logs were acquired while drilling, crossing a sequence of shale and sandstone before reaching the limestone formation. This section begins in the Wara Formation and continues through Mauddud into Burgan Formation (Figure 6).

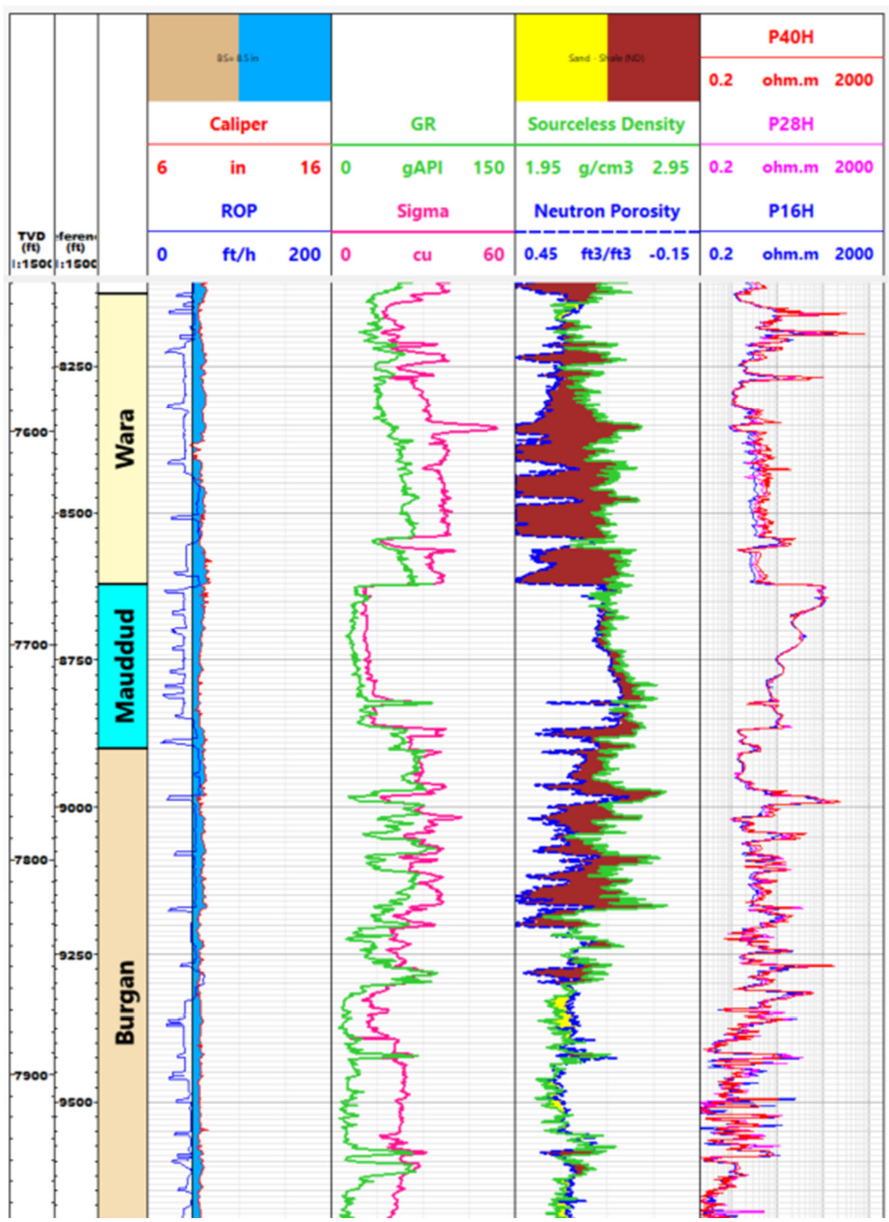

Figure 6—Overview of well 2 illustrating Wara, Mauddud and Burgan formations. Looking at the log data tracks starting from the left track to the right, track 1 provides hole shape indication with caliper and rate of penetration (ROP). The well was drilled at an average 45 ft/ hr. Track 2 shows gamma ray, sigma, track 3 shows the sourceless density and neutron porosity, while track 4 shows the propagation resistivity.

The Wara Formation consists of sandstone and is interbedded with shale. The Mauddud Formation, overlying Burgan is the shallow marine carbonate-dominated succession resulting from the transgression that followed the deposition of the fluvial-deltaic related clastic dominated sediments of the Burgan formation. The Burgan Formation, named after the Burgan field - the largest oil field in Kuwait- is the oldest member of the Wasia group representing the Middle Cretaceous. The Burgan Formation is composed of siliciclastic sediments characterized by well-bedded, well-sorted, rounded, medium- to coarse-grained littoral sands deposited near a delta front on a gradually sinking shelf where carbonates are present. The main target formation for this section was Wara Formation, but the borehole was also drilled into the Burgan formation. After reaching the target depth of this section, the BHA became stuck and could not be retrieved, despite best efforts to free and retrieve it. While this is not a desirable outcome by any means, this incident highlights a significant advantage of using a sourceless BHA/tool. In this case, the days spent on attempting to get the BHA free and abandonment were only 5 days, compared to 15 days for a tool with a chemical radioactive source.

In addition to jarring and acid jobs, which are needed for all types of BHAs that become stuck, a stuck LWD tool with a chemical radioactive source would have required multiple fishing attempts as mandated by regulations. However, for a sourceless BHA, these additional attempts are not required. Furthermore, according to KOC best practices and regulations, if a BHA with a radioactive source is Lost-in-Hole (LIH), the cement top of the plug must be set 500 ft above the fish. In this case, because the BHA was sourceless, this requirement did not apply. As a result, the Wara Formation was later logged with advanced wireline measurements to complete evaluation objectives, and there was sufficient rathole above the fish in the Burgan to achieve this safely. This would not have been possible if the lost-in-hole BHA had radioactive sources.

Despite the LIH issue, data acquisition from the LWD sourceless platform was completed in real-time, delivering the full suite of measurements required to characterize the target reservoirs, as shown in Figure 6.

A correlation with an offset well, located some distance away, was performed to confirm the consistency of reservoir properties. Figure 7 shows the correlation between Well 2, logged with the LWD sourceless logging platform, and a vertical offset well logged using wireline triple combo (GR, resistivity, and density-neutron). The correlation shows a consistent log response despite the lateral heterogeneity.

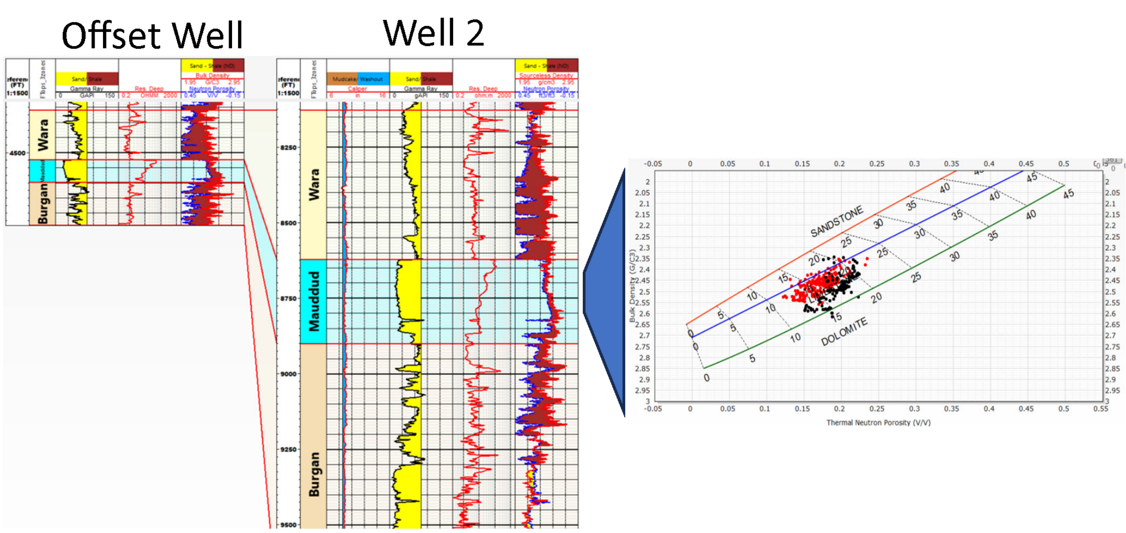

Figure 7—Left -Well correlation between the offset well and Well 2 in Wara, Mauddud and Burgan formation. Right - The density-neutron crossplot shows the Mauddud (limestone) from both wells, with differences attributed to the varying amount of clay.

The density-neutron crossplot from both wells, well 2 (red) and the offset well (black), at Mauddud formation, shows that the limestone in the offset well contains a higher volume of clay compared to Well 2, as indicated by slightly higher GR readings.

Conclusions and Way Forward

The deployment of the sourceless LWD platform in the Abdali and Ahmadi fields successfully enabled the acquisition of high- quality formation evaluation data under challenging drilling conditions. By eliminating the use of radioactive sources, technology significantly reduced Health, Safety, and Environmental (HSE) risks and removed the need for complex abandonment procedures.

This new technology enhanced drilling efficiency and provided real-time data essential for accurate reservoir characterization and economic evaluation of both development and exploration targets. The platform's successful performance in washdown mode further demonstrated its reliability in unstable formations and extended-reach wells with narrow pore pressure margins.

Given this recent successful deployment of sourceless formation evaluation logging while drilling technology, KOC has developed a workflow to guide its strategic utilization. It is designed to ensure that the application of technology stays technically proper and operationally beneficial. KOC will continue to implement this technology under specific conditions, such as:

a. Drilling wells with an elevated risk of stuck pipe or loss of bottom-hole assembly (BHA), where the elimination of radioactive sources significantly reduces the complexity of fishing and abandonment procedures and simplifies the process of regulatory compliance.
b. Drilling in fields located near human habitation, where minimizing the use of hazardous materials is a priority due to associated HSE risks from the radioactive source and considerations of social responsibility.
c. Drilling of development wells where the subsurface is well understood through various offset well data and the primary objective of the current logging run is formation correlation using triple combo measurements.
d. Drilling in remote or logistically complex environments where the handling, transportation, and regulatory compliance associated with radioactive sources present significant operational challenges.

This approach ensures a balance between safety, operational efficiency, and good data quality while enabling the broader adoption of sourceless technology.

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